

Week 5 Operational Diagnostics Report

Oil & Gas Production Throughput Investigation

CONTEXT

A review of 2,920 operational records across four depots was conducted to identify the drivers behind throughput reduction. The investigation evaluated depot performance, machine behaviour, maintenance patterns, incidents, and operational conditions.

INSIGHT

Analysis identified Nairobi depot as the primary area of underperformance. Further drill-down identified NBI-P03 as the dominant contributor. NBI-P03 accounted for approximately 93% of low-throughput events. The machine transitioned from normal production levels in January to a sustained lower-performance state from February onward.

Operational Root Cause

The main driver identified was maintenance-related degradation on NBI-P03. During normal operation, NBI-P03 achieved median throughput of approximately 1,057 barrels. During maintenance-affected periods, median throughput reduced to approximately 626 barrels.

ACTION

1. Prioritise NBI-P03 technical recovery through targeted inspection and corrective maintenance.
2. Strengthen preventive maintenance compliance and overdue maintenance monitoring.
3. Implement throughput monitoring alerts to identify early production degradation.

Executive Conclusion

The production issue is not network-wide. It is concentrated around one asset and one operational failure pattern. Restoring NBI-P03 reliability represents the highest-impact intervention opportunity.

Supporting Evidence Charts



